Digital Signal Processing

Zhou Hang

The Application of FFT Algorithm

※ Using N-point complex sequence FFT, calculate $2N$-point real sequence FFT

※ Using N-point complex sequence FFT to calculate the FFT of $2N$ point real sequence

※ Using N-point complex sequence FFT to calculate the IFFT of N-point complex sequence

Using the N-point complex sequence FFT algorithm, calculate two N-point real sequence FFT

$x_1[k]$ and x_2 [k] are N-point real sequences, how to call the N-point complex sequence FFT algorithm once to obtain X_1[m] and X_2[m]?

Construct N-point complex sequence $y[k] = x_1[k] + \mathrm{j}x_2[k]$

Obtain the DFT of $y[k]$ using the N-point complex sequence FFT algorithm.

Because

$$y[k] = x_1[k] + \mathrm{j}x_2[k] \xleftrightarrow{\text{DFT}} Y[m]$$

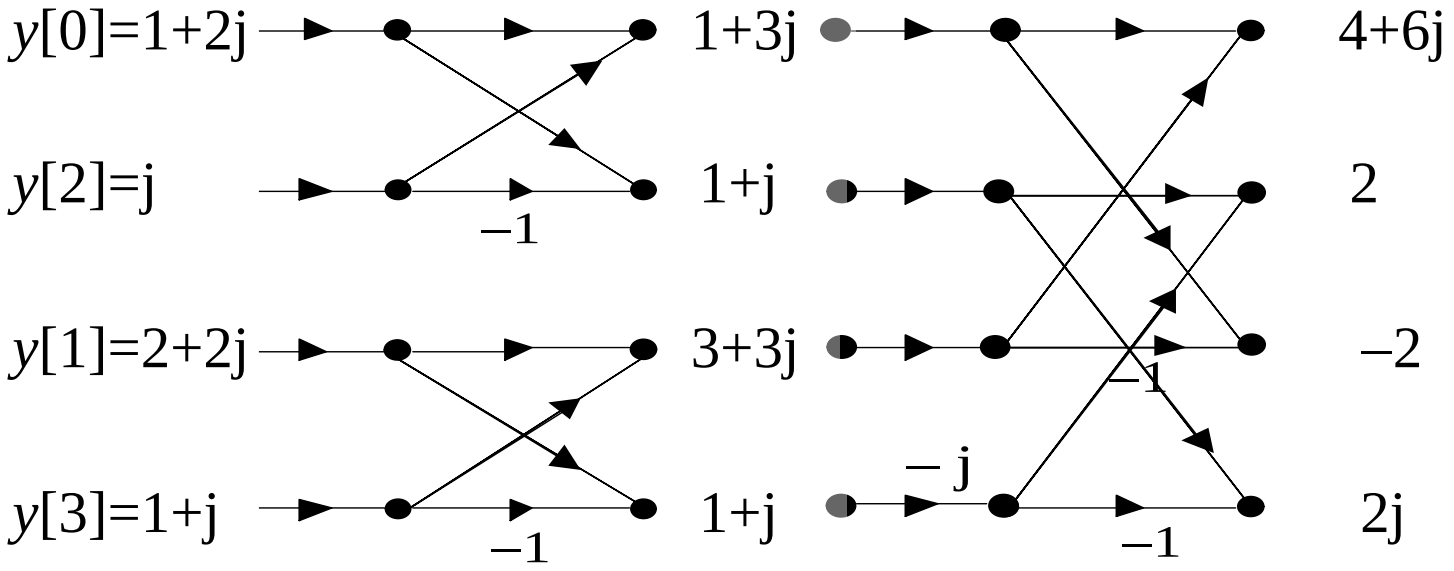
$$y^*[k] = x_1[k] - \mathrm{j}x_2[k] \xleftrightarrow{\text{DFT}} Y^*[(N-m)_N]$$

So

$$X_1[m] = \frac{1}{2}\left\{Y[m] + Y^*[(N-m)_N]\right\} \qquad X_2[m] = \frac{1}{2\mathrm{j}}\left\{Y[m] - Y^*[(N-m)_N]\right\}$$

E.g: Given two 4-point sequences $x[k]=\{1,2,0,1\}$, $h[k]=\{2,2,1,1\}$ Try a 4-point FFT flow chart once and calculate the DFT of two sequences.

Solution: Constructing complex sequences $y[k]= x[k]+ jh[k]=\{1+2j, 2+2j, j, 1+j\}$



E.g: Given two 4-point sequences $x[k]=\{1,2,0,1\}$, $h[k]=\{2,2,1,1\}$ Try a 4-point FFT flow chart once and calculate the DFT of two sequences.

$$\downarrow$$

$$Y[m] = \{4+6j,\quad 2,\ -2,\ 2j\} \qquad m=0,1,2,3$$

$$\downarrow$$

$$Y^*[(4-m)_4] = \{4-6j,\ -2j,\ -2,\ 2\} \qquad m=0,1,2,3$$

$$X[m]=\frac{1}{2}\{Y[m]+Y^*[(4-m)_4]\}=\{\ 4,\ 1-j,\ -2,\ 1+j\ \}$$

$$H[m]=\frac{1}{2j}\{Y[m]-Y^*[(4-m)_4]\}=\{\ 6,\ 1-j,\ 0,\ 1+j\ \}$$

Using the *N*-point complex sequence FFT algorithm to calculate the FFT of a 2*N* point real sequence

X[*k*] is a real sequence of length 2*N*. How to call the *N*-point complex sequence FFT algorithm once to obtain *X*[m]?

$$x[k] \rightarrow \begin{cases} x_1[k] = x[2k] \\ x_2[k] = x[2k+1] \end{cases} \qquad k = 0,1,\mathrm{L}\ ,N-1$$

Constructing complex sequences $y[k]$ $\quad y[k]=x_1[k]+\mathrm{j}x_2[k]$

Calculate $Y[m]$ corresponding to N-point complex sequence $y[k]$, and calculate $X_1[m]$ and $X_2[m]$ from $Y[m]$

Using the *N*-point complex sequence FFT algorithm to calculate the FFT of a 2*N* point real sequence

Synthesize $X[m]$ from $X_1[m]$ and $X_2[m]$ according to base 2

$$\begin{aligned} X[m] &= X_1[m] + W_{2N}^{m} X_2[m] \\ X[m+N] &= X_1[m] - W_{2N}^{m} X_2[m] \end{aligned} \qquad m = 0,1,\mathrm{L}\ ,N-1$$

$$x[k] \rightarrow \begin{cases} x_1[k] = x[2k] \\ x_2[k] = x[2k+1] \end{cases} \qquad k = 0,1,\mathrm{L}\ ,N-1$$

E.g: Calculate the DFT of an 8-point sequence x [k]={1, -1, 1, -1, 2, -1, 1, -1} using an FFT flow graph extracted from a 4-point basis 2 time.

Solution:

According to the principle of the radix-2 time extraction FFT algorithm, the 8-point sequence DFT can be expressed by the DFT of two 4-point sequences. If the sequence $x[k]$ is decomposed into $x_1[k]$ and $x_2[k]$ according to parity, then there exists

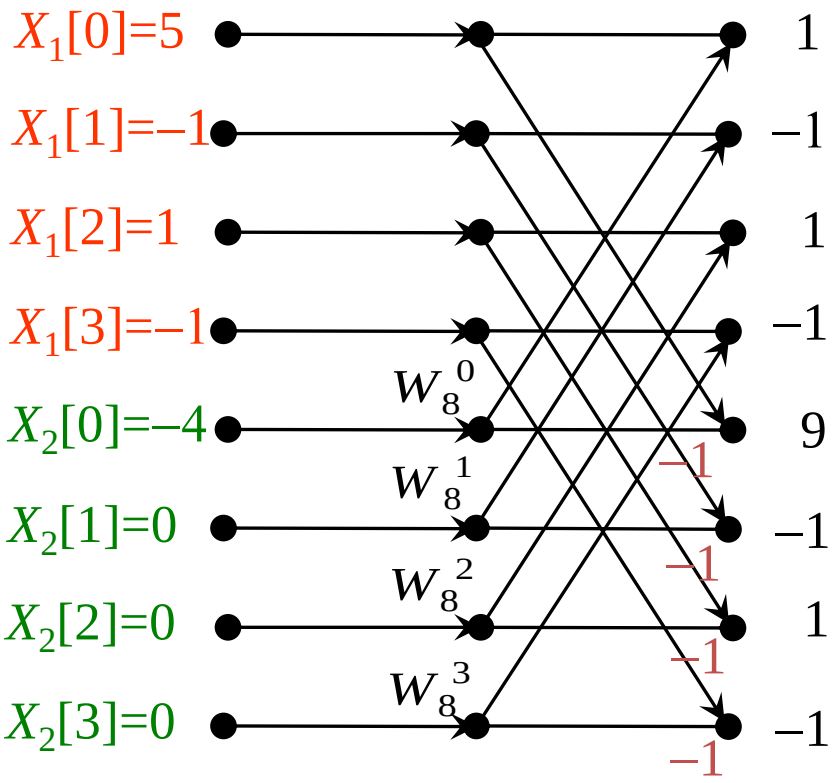
$$\begin{aligned} X[m] &= X_1[m] + W_8^m X_2[m] \\ X[m+4] &= X_1[m] - W_8^m X_2[m] \end{aligned} \qquad m = 0,1,2,3$$

$$x_1[k]=\{1, 1, 2, 1\}$$

$$x_2[k]=\{-1, -1, -1, -1\}$$

Construct a complex sequence $y[k]$ from $x_1[k]$ and $x_2[k]$, calculate $Y[m]$ corresponding to $y[k]$ through 4-point FFT, and calculate $X_1[m]$ and $X_2[m]$ from $Y[m]$.

E.g: Calculate the DFT of an 8-point sequence x [k]={1, -1, 1, -1, 2, -1, 1, -1} using an FFT flow graph extracted from a 4-point basis 2 time.



$X_1[m]=\{5, -1, 1, -1\}$

$X_2[m]=\{-4, 0, 0, 0\}$

Then

$X[m]= \{1,-1,1, -1, 9, -1, 1,-1\}$

Using N-point complex sequence FFT to calculate the IFFT of N-point complex sequence

Definition:

$$X[m] = \mathrm{DFT}\{x[k]\} = \sum_{k=0}^{N-1} x[k] W_N^{mk}$$

$$x[k] = \mathrm{IDFT}\{X[m]\} = \frac{1}{N} \sum_{m=0}^{N-1} X[m] W_N^{-mk}$$

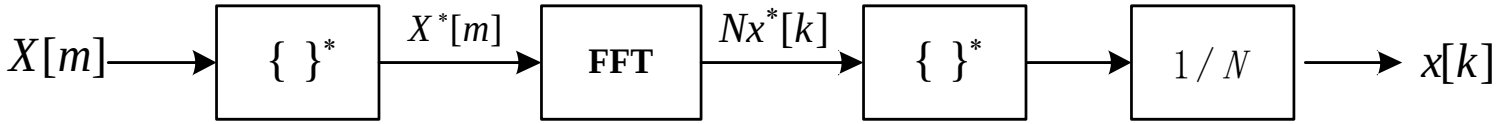
Symmetry defined by DFT and IDFT

$$x[k] = \mathrm{IDFT}\{X[m]\} = \frac{1}{N} \sum_{m=0}^{N-1} X[m] W_N^{-mk}$$

$$= \frac{1}{N} \left(\sum_{m=0}^{N-1} X^*[m] W_N^{mk} \right)^* = \frac{1}{N} \left(\mathrm{DFT}\{X^*[m]\} \right)^*$$

Using N-point complex sequence FFT to calculate the IFFT of N-point complex sequence

$$x[k]=\frac{1}{N}\left(\sum_{m=0}^{N-1}X^*[m]W_N^{mk}\right)^*=\frac{1}{N}\left(\mathrm{DFT}\{X^*[m]\}\right)^*$$



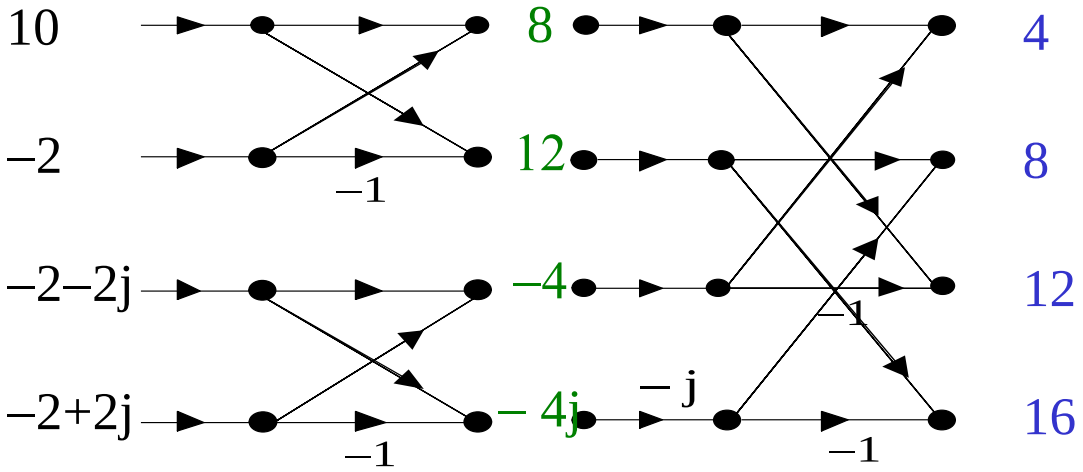
Step: (1) Select the conjugate of $X[m]$

(2) Calculate using FFT flow graph $\mathrm{DFT}\{X^*[m]\}$

(3) Take the conjugate of the result in (2) and divide it by N

E.g: Given $X[m]$=DFT $\{x[k]\}$=$\{10, -2+2j, -2, -2-2j\}$,extract FFT stream graph using base 2 time and calculate $x[k]$.

$$X^*[m]=\{10,-2-2j,-2,-2+2j\} \qquad \text{DFT}\left\{X^*[m]\right\}=\{4, 8, 12, 16\}$$



$$x[k]=\frac{1}{4}\left\{\text{DFT}\left\{X^*[m]\right\}^*\right\}=\{1, 2, 3, 4\}$$

The Application of FFT Algorithm

MATLAB provides multiple functions for fast computation of DFT and IDFT, namely

fft(x)， fft(x, N)， ifft(X)， ifft(X, N)

FFT (x) calculates the DFT of point L, where L is the length of sequence x, i.e. L=length (x);

FFT (x, N) calculates the DFT of N points. If L<N, the original sequence is zeroed to N points,

Then calculate its *N*-point DFT;

IFFT (X) calculates the IDFT of point L, where L is the length of sequence X;

IFFT (X, N) calculates the IDFT of N points. If L<N, the original sequence is zeroed to N points.

Given that $X[m]$ is the DFT of a $2N$ point real sequence $x[k]$, briefly describe the process of calling an N-point FFT program once to calculate the IDFT $x[k]$ corresponding to $X[m]$.

If $x_1[k]$ is an even component of $x[k]$, and $x_2[k]$ is an odd component of $x[k]$

The DFT of $x_1[k]$ is $X_1[m]$, and the DFT of $x_2[k]$ is $X_2[m]$, then we have:

$$X[m] = X_1[m] + W_{2N}^{m} X_2[m]$$

$$X[m+N] = X_1[m] - W_{2N}^{m} X_2[m]$$

From the above equation, it can be concluded that:

$$X_1[m] = \frac{1}{2}\{X[m] + X[m+N]\} \qquad X_2[m] = \frac{1}{2} W_{2N}^{-m}\{X[m] - X[m+N]\}$$

Given that $X[m]$ is the DFT of a $2N$ point real sequence $x[k]$, briefly describe the process of calling an N-point FFT program once to calculate the IDFT $x[k]$ corresponding to $X[m]$.

(1) Given $X[m]$, first obtain $X_1[m]$ and $X_2[m]$:

$$X_1[m]=\frac{1}{2}\{X[m]+X[m+N]\} \qquad X_2[m]=\frac{1}{2}W_{2N}^{-m}\{X[m]-X[m+N]\}$$

(2) Let $Y[m]=X_1[m]+\mathrm{j}X_2[m]$, and based on calling an N-point FFT once to implement IFFT, obtain $y[k]$ corresponding to $Y[m]$.

(3) Obtain $x_1[k]$ and $x_2[k]$ from $y[k]$:

$$x_1[k]=\mathrm{Re}\{y[k]\}$$
$$x_2[k]=\mathrm{Im}\{y[k]\}$$

Given that $X[m]$ is the DFT of a $2N$ point real sequence $x[k]$, briefly describe the process of calling an N-point FFT program once to calculate the IDFT $x[k]$ corresponding to $X[m]$.

Obtain $x[k]$ corresponding to $X[m]$ from $x_1[k]$ and $x_2[k]$:

$X_1[k]$ is the even component of $x[k]$

$X_2[k]$ is the odd component of $x[k]$

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.